

TECHNICAL SOLUTION INTENT

Customer 360

MCP fleet architecture for the Commercial Credit Brain

The Commercial Credit Brain is the orchestrating intelligence; the MCP fleet is the tool layer it navigates, built one server at a time rather than big-bang. The shipped credit-memo tools (Boom, AFS, experience-mcp) were the first set, deal-level underwriting. This Customer 360 / Early-Warning fleet is the next, relationship-level monitoring. Each server that comes online widens what the brain can reach, so the brain gets smarter every time a tool ships. Policy and governance are designed in from day one, not bolted on: the SR 11-7 deterministic wall, run-as-user identity, per-call audit, DRAFT-until-attested. And the fleet is enterprise-bank scalable on AWS, remote-hosted in org-controlled infrastructure so untrusted, high-churn MCP code never touches an analyst's machine.

SURFACE	SCALE	STAGE
One MCP server, many swappable surfaces: Claude MCP App widget, Agentforce-in-FSC twin, hosted Next.js, M365 / Copilot, ChatGPT Apps. Built to the strict MCP Apps subset, portable across hosts.	Remote-host the fleet in org-controlled AWS infra (Bedrock AgentCore Runtime or ECS / Fargate in a private VPC). The governance unit becomes the environment plus a promotion gate.	Stage 0 reference build. Inter-MCP fleet pattern shipped and proven in code; Customer 360 read spine in build. Honest gates stated, not hidden.

01 The through-line, and the 4-tier fleet model

SPINE

The Commercial Credit Brain is the orchestrating intelligence. The MCP fleet is the tool layer it navigates, and we build that layer one server at a time, so the brain's reach widens with every server that ships.

The hard-won architectural principle, proven on the shipped credit-memo fleet: **reasoning and methodology live in the plugin; tools and data live in the connectors**. No duplication across the seam. The brain does not change when a tool is added, it just discovers more tools. Reasoning stays in the plugin; capability arrives as data and tools.

T	COMPONENT	HOLDS	IDENTITY / TRUST
1	Reasoning / brain Agents · Agentforce · Cowork (MCP client)	Methodology, orchestration, ranking, narration. Never computes a regulated number.	The model is downstream of the math.
2	SF-native Customer 360 MCP nCino LLC_BI_* + FSC FinServ_*, one org	The relationship read surface.	Run-as-user OAuth (PKCE, mcp_api + refresh_token) → WITH USER_MODE so native FLS / sharing apply. Gated admin god-mode is the one audited bypass.
3	Aggregation / experience MCP experience-mcp	Cross-source fetch + join-in-code, plus the decision ledger / audit (Snowflake).	Itself an MCP client to the system MCPs; write path = client-credentials integration user, actorStamp() audit, EXPERIENCE_ALLOW_WRITES gate.
4	System MCPs Boom · AFS · IRIS · CapIQ / IBIS	Raw financial / market data plus deterministic compute (spread, ratios, PD, peer medians).	Each a standalone connector behind the aggregation tier.

THE SR 11-7 WALL, STATED AS A CALL GRAPH

The **deterministic** edges into experience-mcp and the system MCPs are the only edges that *produce* a regulated number: covenant.js computes grade = Boom ratio (actual) × nCino threshold (definition) as a pure function; boomFinancials.js retrieves a Boom-computed ratio and the number never passes through the LLM. Every edge into the reasoning client is **narrate**: the LLM ranks and explains values that were already computed in code. "Agents reason, tools compute" is not a slogan here, it is the literal call graph.

02 The incremental build: deal-level to relationship-level to beyond

ROADMAP

We build the fleet one server at a time. The brain's reach widens with every wave, and each new server inherits the policy fence rather than inventing its own.

WAVE	FLEET ADDED	WHAT THE BRAIN CAN NOW DO	STATUS
1 · Deal-level underwriting	Boom (spreading) · AFS (servicing) · experience-mcp (compose + nCino write)	Draft and write back a covenant-graded commercial credit memo on one deal.	SHIPPED
2 · Relationship-level monitoring	Customer 360 SF-native MCP (run-as-user nCino + FSC) · IRIS · CapIQ / IBIS · Early Warning	Move from one deal to the whole obligor / household: live portfolio, share-of-wallet, early-warning signals.	IN BUILD
3+ · Beyond	Next source servers (peers, market, treasury, ...)	Each new server is a new sense for the same brain.	ROADMAP

Why incremental is also safe

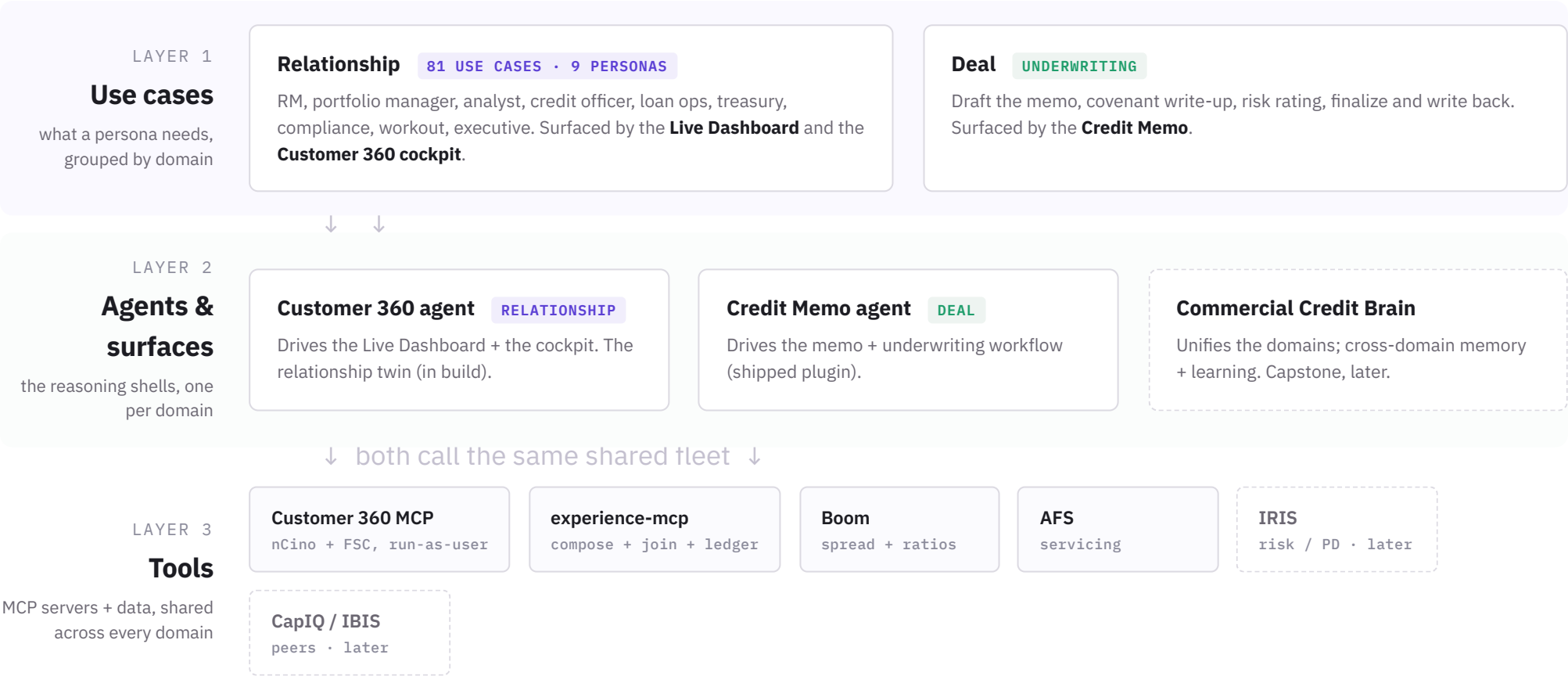
- Each MCP server is an **independent unit** (own repo, own deploy, own scope). Adding one cannot regress the others.
- The brain does not change when a tool is added; it just discovers more tools. Reasoning stays in the plugin, capability arrives as data.
- **Policy and governance are designed in from day one:** SR 11-7 deterministic wall, run-as-user identity, per-call audit, DRAFT-until-attested. A new server inherits the fence; it does not get to invent its own.

VERSION	CAPABILITY ADDED	GOVERNANCE POSTURE	STATUS
v1 · Read-only	Customer 360 run-as-user reads (nCino + FSC); cross-source compose; covenant-grade join.	Run-as-user OAuth + <code>USER_MODE</code> ; SR 11-7 wall; DRAFT-until-attested.	Spine <code>SHIPPED</code> , reads in build
v2 · Write path	nCino nFORMS publish + section sync; AFS workpackage staging.	Client-credentials integration user, <code>actorStamp()</code> , <code>EXPERIENCE_ALLOW_WRITES</code> gate, Snowflake ledger.	Memo write <code>SHIPPED</code> ; C360 next
v3 · Live sources	IRIS / Snowflake (rating, PD, covenant grade) + CapIQ / IBIS peers wired live; Early Warning thresholds.	Swap-later grade reads the published IRIS grade; deterministic EWS triggers.	ROADMAP
v4 · Scale	Remote-host the fleet in org-controlled AWS infra; promotion-gated environments.	The environment becomes the governance unit; owned egress + IAM.	ROADMAP

03 The three layers: agents, tools, use cases

ORIENTATION

These are three different axes, not one. Use cases belong to a domain. Agents are the reasoning shells, one per domain. Tools are the MCP servers, shared across domains. Read top down: a persona's need is met by a surface, which calls the shared fleet.



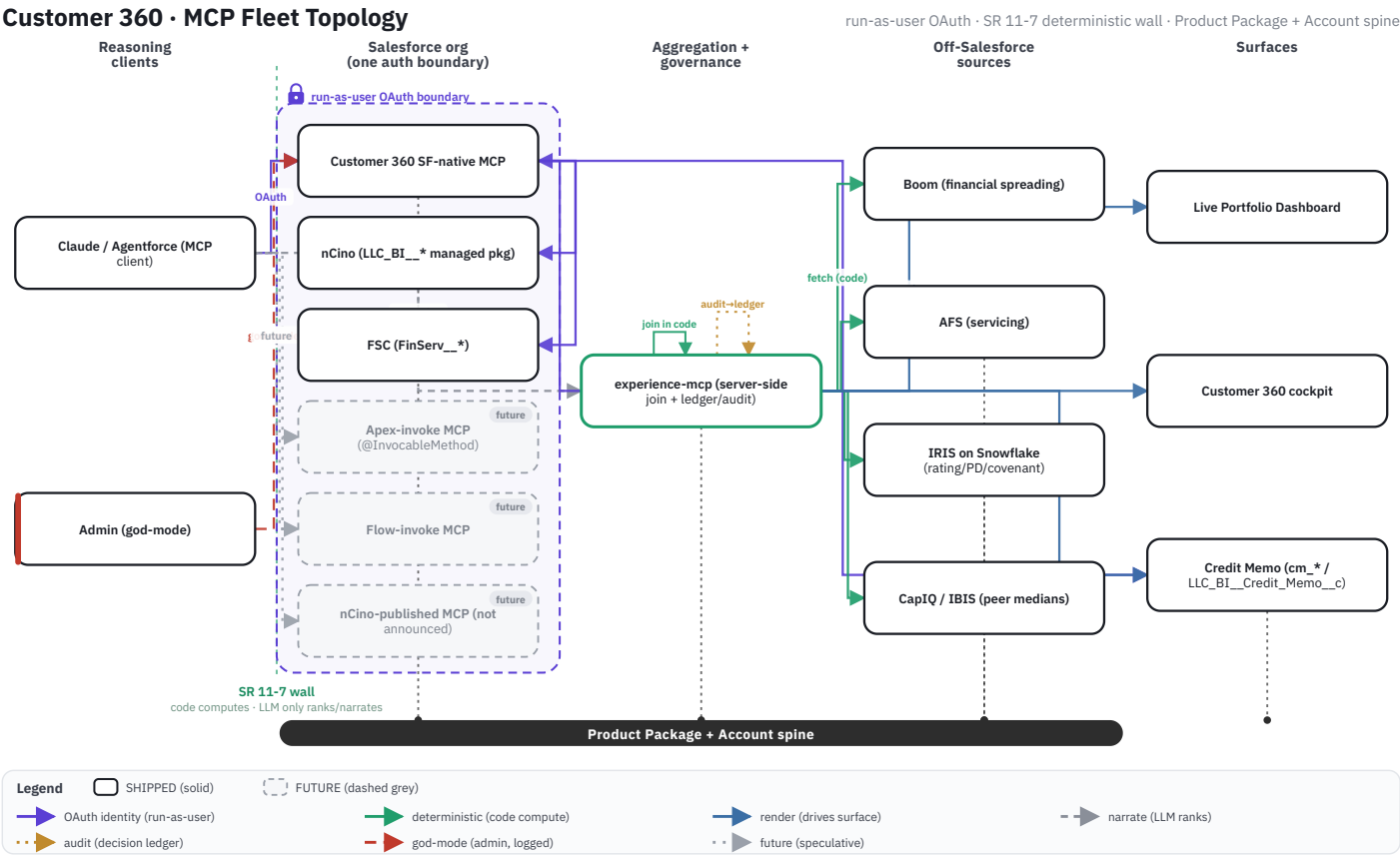
Use cases belong to a domain (relationship vs deal). **Agents** are per-domain shells; the Brain unifies them last. **Tools** are shared: one server serves many use cases across domains.

The 81 use cases are Customer 360. They live inside the Live Dashboard and the cockpit, not as a tool of their own.

04 The MCP fleet topology

ARCHITECTURE

Five bands on one spine: reasoning clients → a single Salesforce org auth boundary → the aggregation / governance tier → off-Salesforce sources → the surfaces that render off the joined, governed result. The Product Package + Account pair is the rail every node keys on.



FLEET TOPOLOGY. REASONING CLIENTS CALL THE CUSTOMER 360 MCP (RUN-AS-USER OAUTH READING NCINO LLC_BI_* + FSC FINSERV_* IN ONE ORG); EXPERIENCE-MCP IS THE JOIN + GOVERNANCE TIER AND IS ITSELF AN MCP CLIENT TO THE OFF-SALESFORCE SOURCES (BOOM, AFS, IRIS / SNOWFLAKE, CAPIQ); THREE SURFACES (LIVE PORTFOLIO DASHBOARD, CUSTOMER 360 COCKPIT, CREDIT MEMO) RENDER OFF THE RESULT. DOTTED NODES (APEX_MCP, FLOW_MCP, NCINO_MCP) ARE HONESTLY MARKED FUTURE; NCINO PUBLISHES AN AOS AGENT, NOT AN MCP SERVER.

05 How a tool call actually resolves

MECHANICS · 1 OF 2

One banker question becomes one tool call. The fleet composes the answer in code, and the regulated number is computed before the model ever sees it. The aggregation MCP is itself a client to the system MCPs.

- 1

The banker asks REASONING CLIENT · CLAUDE / AGENTFORCE

A single tool call goes to the Customer 360 MCP: `tools/call customer360_show_ews { accountId }`. The model orchestrates; it does not fetch or compute.
- 2

Read, as the banker CUSTOMER 360 MCP · NCINO + FSC

The server authenticates **run-as-user** (OAuth, `WITH USER_MODE`) and reads the relationship, exposures and covenant thresholds. Native Salesforce sharing and FLS decide what this banker may see.
- 3

Fan out, in parallel AGGREGATION TIER, ACTING AS AN MCP CLIENT

It calls the system MCPs at the same time over streamable-HTTP (`tools/call` , 8s timeout, graceful-null so one slow source never blocks the view):

Boom

spread & ratios

`boom_get_ratios`

AFS

servicing behaviour

`revolver_utilization`

IRIS

risk migration

`iris_get_signal`
- 4

Join and compute, in code AGGREGATION TIER · PRODUCT PACKAGE + ACCOUNT SPINE

Results are joined on the one spine and the regulated number is formed here: `ratio × threshold → pass / watch / breach`. **The number is born in code, not in the model.**

SR 11-7 WALL ----- ABOVE: CODE COMPUTES · BELOW: MODEL ONLY NARRATES
- 5

Return a small result BACK TO THE REASONING CLIENT

Only the computed grades and KPIs travel back, not the raw spread. The payload is small and already final.
- 6

Rank and narrate REASONING CLIENT

The model ranks the fired signals and explains them in plain language. **It never recomputes the number** it was handed.
- 7

Commit, gated LATER PHASE

If a human commits a decision, a write goes back to nCino and an audit record to the Snowflake ledger. Reads are v1; writes come later, behind the gate.

Pattern verified in experience-mcp/src/boomFinancials.js, the aggregation MCP is itself a client to the system MCPs (JSON-RPC tools/call over streamable-HTTP).

06 The inter-MCP mechanics, in shipped code

MECHANICS · 2 OF 2

This is real, shipped code, not a slideware diagram. The cross-source call lives in `experience-mcp/src/boomFinancials.js`.

1 Client → experience / C360 tool.

Claude / Agentforce calls a `tools/call`.

2 Fan-out, server-to-server.

`experience-mcp` `fetch()`s each system MCP over streamable-HTTP with a JSON-RPC envelope, in parallel (`Promise.all`), an 8s timeout (`AbortSignal.timeout(8000)`), graceful-null on any failure. It parses the SSE `data:` frames to pull `result.content[0].text`.

3 Join + compute in code

on the Product Package + Account keys. The regulated number is formed here (`covenant.js gradeCovenant()` → `pass | watch | breach`, 10% watch band), in code, before the LLM ever sees it.

4 Return a small, already-computed result.

The LLM only ranks and narrates.

5 Write-back (optional, gated).

`ncino_publish_credit_memo / ncino_sync_memo_sections` → nCino nFORMS HTML-native memo, behind `EXPERIENCE_ALLOW_WRITES`, run as the right identity, audited to the Snowflake ledger.

```
// experience-mcp/src/boomFinancials.js – the aggregation MCP acting as a CLIENT
const res = await fetch(BOOM_MCP, {
  method: "POST",
  headers: { "Content-Type": "application/json", Accept: "application/json, text/event-stream" },
  body: JSON.stringify({ jsonrpc: "2.0", id: 1, method: "tools/call",
    params: { name, arguments: args } }),
  signal: AbortSignal.timeout(8000), // parallel fan-out, graceful-null
});
// "The number is computed in the Boom MCP, retrieved here server-side,
// and never passes through the LLM."
```

THE LITERAL SEAM, LOAD-BEARING: THIS IS THE SR 11-7 WALL AS CODE.

SWAP-LATER, DESIGNED IN

`covenant.js` computes the grade deterministically today; the moment IRIS-in-Snowflake publishes its own covenant grade, the same tool **reads the published grade instead of recomputing**, a config / env flip, not a rewrite. The fence stays; the source upgrades.

07 Transport, hosting, and why detection is a job

RUNTIME

All three shipped connectors are Next.js + `mcp-handler`, dual-transport stdio + streamable-HTTP. Stdio for local / embedded use; streamable-HTTP for the remote governed connector URL that Cowork / Claude / Agentforce attach to.

- **Stateless connectors, stable resource URIs.** Widgets are served at a content-hash-free `ui://<bank>/<name>.html` URI so the host caches the widget at the connector level. Cross-source widgets live with whoever owns the join (experience-mcp), because external iframes are blocked by the host, so the fleet renders its own bank-branded UI.
- **Public-but-governed URLs.** Today: SSO / deployment-protection off so the connector URL is reachable, writes gated behind `EXPERIENCE_ALLOW_WRITES`, run as a real Salesforce user (client-credentials External Client App). The connector is bank-neutral: the bank name is a `BANK` config block, env-swappable per tenant.

ASYNC + GOVERNOR LIMITS: DETECTION IS A JOB, THE SCREEN JUST READS

Computing an EWS read live (AFS servicing + nCino covenants + FSC deposits + Boom ratios) inside a single realtime Salesforce response would blow **SOQL and CPU governor limits** the moment you fan it across a portfolio. So the architecture splits compute from render: **EWS detection runs as a Queueable / Batch Apex job**, server-side, on the deterministic reads, and writes pre-computed results (fired signals, cushion %, severity score) back to records. The dashboard **reads those pre-computed fields**; no heavy join at view time, so the portfolio view scales to the whole book.

CDC repaints hands-free. Change Data Capture (the shipped `CreditMemoCDC__chn` channel over `LLC_BI__Product_Package__ChangeEvent` + `LLC_BI__Loan__ChangeEvent`) drives an `lightning/empApi` LWC that calls `getRecordNotifyChange` + `RefreshEvent`: the screen moves on a write without a manual refresh (verified live). One pre-demo ritual: warm the CDC channel (~1-3s warm vs ~15-30s cold).

The fence holds in async too. A fired trigger raises a flag; it never moves a grade. EWS detects, grades, ranks, and routes; a human classifies and decides (Pass / Watch / Special Mention / Substandard / Doubtful / Loss). Severity (Low / Med / High / Critical) is decision-support, captured proposed-not-committed; it becomes real only when the authorized human commits via `record_decision`, with classification authority enforced server-side against the acting user.

08 The data spine, the Snowflake ledger, and per-call PII audit

DATA

What makes the joins valid and the decisions defensible: one spine every node keys on, end-to-end lineage, and a decision ledger that records why every number was committed.

One spine: Product Package + Account. Every node keys on this pair. The Account is the relationship anchor (FSC `FinServ__*` + nCino `LLC_BI__Account`); the Product Package is the deal anchor that nCino, the credit memo (`LLC_BI__Credit_Memo__c` / the `cm_*` rich-text fields), AFS servicing, and IRIS grades all hang off. experience-mcp's joins are valid **only because both sides resolve to the same keys**. A fired EWS signal rolls up to the obligor / relationship group, never an isolated loan tile.

Lineage you can trace end to end: Boom → IRIS → nCino. The raw spread is computed in Boom; the rating / PD / covenant grade is published by IRIS on Snowflake (zero-copy); the governed result is written back to nCino as structured `cm_*` narrative fields plus the nFORMS HTML memo, DRAFT until finalized and attested, then one deliberate end-sync commit, never per-edit write-through.

The Snowflake decision ledger + per-call PII audit. experience-mcp owns the system-of-record for *why* a decision was made. Every tool call that touches borrower data writes an audit row: the run-as integration user EXECUTES, the human RM is STAMPED (`actingUserName` / `actingUserId`), with agent-provenance (model + version) as the future third leg of the audit triple. `record_decision` / `log_audit_event` / `recall_decisions` are the ledger surface.

THE WHOLE POSTURE, IN ONE LINE

Inside the org, queries run `WITH USER_MODE` so CRUD / FLS / record-sharing auto-apply to the caller; no anonymous service account, the caller lands in the audit trail. The single admin god-mode path (cross-user read) is explicitly enabled and fully logged. That is the whole posture: **the regulated number is computed in code, the LLM never computes it, and every borrower-data touch is attributable.**

09 Identity and auth: two legs of provenance, run-as-user on reads

IDENTITY

The read path is an upgrade over the memo, not a copy of it. The shipped credit memo (experience-mcp / [sfClient.js](#)) runs as a single client-credentials service account, where [actingUser](#) is an audit stamp on writes only, never a sharing identity. Customer 360 deliberately upgrades this: it runs as the user.

- **Reads, run-as-user OAuth.** External Client App, OAuth 2.1 + PKCE, [mcp_api](#) + [refresh_token](#) scopes; every SOQL / Apex path runs [WITH USER_MODE](#) so CRUD / FLS / record-sharing auto-apply to the caller. No anonymous service account; the calling user lands in the audit trail. Correct for a relationship cockpit that reads far more of the customer than a single deal does.
- **Writes, the SR 11-7 two-leg provenance.** The gated write path uses a client-credentials run-as integration user that EXECUTES, with the acting user STAMPED ([actingUserId](#) / [actingUserName](#), threaded on every write per the plugin↔MCP contract). The agent model + version is the third leg. Two legs of identity plus one of model = full provenance on every committed number.
- **Admin god-mode** is a separate, explicitly-enabled, fully-logged cross-user read path, the one place the run-as-user boundary is intentionally bypassed. Permissioned plus audited.
- **Per-user token isolation must be verified before any borrower-PII pilot.** On custom / Team connectors, confirm the OAuth token is issued per user, not shared per org; a shared token would mean every user sees every book.

HONEST CAVEAT, LIVE-VERIFIED 2026-06-28

In the bankingpt sandbox the internal OWD is **permissive** (ReadWrite Public on most objects), so run-as-user does **not** narrow visibility in this org; persona book-narrowing is UX, not a security boundary here. Real least-privilege enforcement comes from the bank's production OWD (Private + Grant Access Using Hierarchies). We present persona views; we do not claim we enforce least privilege in the sandbox.

10 Enterprise scale on AWS: the environment is the governance unit

SCALE

Threat model, stated plainly: an MCP server is arbitrary code running as the user. It must never run on an analyst's machine. Remote-host the fleet in org-controlled infrastructure.

LANDING PATTERN A

Bedrock AgentCore Runtime

Per-session microVM, isolated then terminated after use. Accenture Track 2 is already standing up an AgentCore AWS environment, so the v4 target has a concrete home.

LANDING PATTERN B

ECS / Fargate in a private VPC

The fleet runs behind owned egress and IAM. Source MCPs (Boom, AFS, IRIS, CapIQ) hold no session state, so they scale out flat behind the per-session microVMs.

- **The governance shift.** The unit of control becomes the **environment + a promotion gate**, not per-connector approval. You don't vet each connector by hand; you vet the environment it's promoted into. This is what makes untrusted, high-churn MCP code safe at bank scale.
- **Residual risk relocates to ownable surfaces:** VPC egress + IAM, both owned by the bank, both auditable.
- **The SR 11-7 wall holds across all of it.** Every regulated number is computed deterministically in code (`covenant.js` grade = actual × threshold; `boomFinancials.js` server-to-server fetch, the ratio never round-trips through the LLM). The decision ledger and audit trail live in Snowflake (`record_decision` / `log_audit_event`). DRAFT-until-attested; the cockpit concludes and routes, a human commits.

THE V4 ARGUMENT, IN ONE FRAME

Remote-host the fleet in org-controlled infrastructure so untrusted, high-churn MCP code never touches an analyst's machine. The governance unit becomes the environment plus a promotion gate; egress and IAM are owned by the bank, not the tool author. This is not hypothetical: Accenture Track 2 is already standing up an AgentCore AWS environment.

11 Bank-agnostic by construction

CONFIG

One build, every bank, via config, not rework. A managed-package namespace plus a per-tenant config map, no code fork per institution.

CONFIG SEAM	WHAT IT SELECTS	NAMED DEFAULT
groupingKey	Relationship / household grouping.	The nCino Connection graph (208 edges, the populated authoritative ownership model).
Covenant / collateral generation	Legacy-vs-modern object selection.	Covenant2 / Loan_Collateral2 vs the empty legacy Covenant__c / Loan_Collateral__c .
walletSource	Wallet priority.	nCino-commercial-first, then FSC, unioned for full share-of-wallet.
EWS thresholds + SLAs + scorecard weights	The "tomorrow" detail.	Every threshold in EARLY-WARNING.md is a labeled default the bank overrides by config.

THE PAYOFF

A bank with seeded nCino treasury gets the commercial wallet automatically, **no code change, just config priority**. The covenant / collateral seam, the wallet union, and every EWS threshold ship as named, replaceable defaults; a new institution drops in by config, not by rework.

This is what makes the incremental fleet portable: each new server inherits the same per-tenant config map and the same managed-package namespace, so widening the brain's reach never means re-forking the build for the next bank.

12 One server, many swappable surfaces

SURFACES

The render layer is a distribution choice, not an auth-feasibility one. There is exactly one thing to build and harden: the Customer 360 MCP server. Every surface bolts onto it; the data plane, per-user authorization, and FLS / sharing enforcement are identical across all of them.

SURFACE	AS-USER AUTH	LIVE REFRESH	SEC BURDEN	DEMO-READY NOW
Claude chat / desktop MCP App widget	OAuth 2.1 + PKCE, user-scoped token → SF ECA; <code>USER_MODE</code>	Yes (<code>callServerTool</code> re-call, true live)	Med	Yes, we self-host the <code>ui://</code> widget
Agentforce in FSC CLT / LWC twin	Native run-as-user; identity IS the SF session, no bridge	Yes (tool re-call, native session)	Lowest	Yes, CLT-in-Lightning shipping
M365 / Copilot MCP App widget	Entra OAuth 2.1 / SSO + Entra→SF on-behalf-of exchange	Yes (same MCP Apps bridge)	Med + the bridge	GA Apr 7 2026
Hosted Next.js per-user SF OAuth	Per-user SF OAuth auth-code + PKCE; all SOQL / Apex as that user	Yes (SSR / streaming)	High	Yes, but "another tab"
ChatGPT Apps Apps SDK / MCP	OAuth 2.1 auth-code + PKCE → SF ECA; <code>USER_MODE</code>	Yes (same bridge)	Med + submission	Yes (Business / Enterprise beta)

PORTABILITY, AND THE HONEST DEAD ENDS

Build the widget to the strict MCP Apps subset and one `ui://` widget is portable across Claude, ChatGPT, and Copilot; surfaces become a menu we light up in sequence, not parallel rebuilds. Honest caveats: server-side EWS detection is a Salesforce Apex / Queueable job the dashboard **reads**, so no chat surface does background monitoring, all render fresh-on-view; "share one link, everyone sees it live" is not how any surface works, and a bank does not want it to, each viewer authorizes their own identity; Claude Code artifacts are a dead end (static, no network, cannot reach the server).

13 Policy and governance designed in from day one

POSTURE

What a bank infosec / AppExchange review will check, and the control that answers it. The single most important line: SR 11-7 deterministic compute happens in experience-mcp / source servers; the LLM only ranks and narrates. No regulated number is produced by the model.

CONTROL	MECHANISM	REVIEW QUESTION
Run-as-user identity	Customer 360 runs as the caller via per-user OAuth (ECA, PKCE, <code>mcp_api</code> + <code>refresh_token</code>). No anonymous service account on the read spine.	Whose data can this identity see, and is sharing platform-enforced?
WITH USER_MODE	Every Apex / SOQL read runs <code>WITH USER_MODE</code> so CRUD / FLS / record-sharing auto-apply to the caller, platform-enforced, not re-coded.	Is FLS / record-sharing enforced on every query, including aggregations?
Per-user token isolation	Tokens issued per user, not shared per org. Sharp edge flagged honestly: on custom / Team connectors this must be verified before any borrower-PII pilot.	Is the OAuth token issued per user or shared per org?
Per-call PII audit + ledger	experience-mcp owns the Snowflake decision ledger + audit trail: <code>log_audit_event</code> on every fired signal, <code>record_decision</code> when a human commits.	Can every read and decision be traced to a user and a timestamp?
SR 11-7 deterministic wall	The regulated number (Boom ratio, <code>deal_covenant_grade</code> , AFS utilization, IRIS PD) is computed in code server-side and never round-trips through the LLM.	Does the model produce any regulated figure or make any credit decision?
DRAFT-until-attested	EWS detects, grades, ranks, routes; a human classifies and decides. The Borrower Health Index (0-100) is decision-support and maps to but never overrides <code>LLC_BI__Risk_Rating__c</code> .	Can the system commit a grade, classification, or write autonomously?
Admin god-mode seam	A separate, explicitly-enabled, fully-logged cross-user read path (kind "god-mode") for internal + demo, behind an explicit switch + permission.	Is there any path that bypasses run-as-user, and is it gated + logged?

HONEST SHARP EDGE

In the bankinggpt sandbox the internal sharing model is permissive (ReadWrite Public on most objects), so run-as-user does **not** narrow visibility there; persona book-narrowing is UX, not a security boundary in that org. Real least-privilege enforcement needs the bank's production OWD (Private + Grant Access Using Hierarchies) or server-side acting-user scope filters. The cockpit presents persona views but must not claim it enforces least privilege in this org. OWD verification per object is an open item.

14 Status: what is built vs what is needed

STATUS

Grounded strictly in shipped code and the verified sandbox. Shipped solid, future dashed, no aspirational architecture.

SHIPPED	NEEDED
Inter-MCP fleet pattern (the proof) experience-mcp is itself a client to the system MCPs: <code>boomFinancials.js</code> JSON-RPC <code>tools/call</code> over streamable-HTTP, SSE-parsed, 8s timeout, graceful-null; <code>covenant.js</code> deterministic ratio × threshold join; <code>route.js</code> dual-transport <code>mcp-handler</code> with bank-branded <code>ui://</code> widgets. Shipped source MCPs Boom (spreading + ratios), AFS (<code>revolver_utilization</code>, <code>payment_history</code>, <code>loan_summary</code>, <code>create_workpackage</code>), IRIS on Snowflake (node shipped, live wiring stubbed), experience-mcp (join + Snowflake ledger + audit). Verified nCino + FSC schema Piedmont real C&I shape: Operating Company + 100%-owner individual full-note guaranty, 4 <code>Covenant2</code> (all Compliant), 5 <code>Loan_Collateral2</code> (~\$10M lendable), Product Package TCE \$12.5M grade 5, Connection graph 208 edges. Fetch-ready now (no new build) nCino + FSC reads, Boom ratios / spread, AFS servicing, <code>deal_covenant_grade</code>, the Snowflake ledger (<code>recall_decisions</code>): covers identity / scoping, the KPI tiles, the borrower-health rail, single-customer 360, and 5 of 7 EWS signal classes.	Native Customer 360 MCP read spine TO BUILD The run-as-user nCino + FSC read spine itself (we proxy via subject-sf today). NEXT-STEPS Task A. Live IRIS / Snowflake + CapIQ / IBIS STUBBED 2 of 7 EWS signal classes (risk migration; external / industry peers) are placeholder-seeded until the rating / PD and peer-median wiring is live. The EWS async detection job TO BUILD Server-side overnight detection is a Salesforce Apex / Queueable job the dashboard reads. Triggers + config seams framed in EARLY-WARNING.md; the job is unbuilt. Bank-specific EWS config INCOMING Thresholds, escalation SLAs, scorecard weights, and the EWS-severity → bank-watch-grade mapping. Every threshold ships as a named, replaceable default; drops in by config. Token isolation + production OWD VERIFY Confirm per-user tokens (not shared per org) before any borrower-PII pilot; confirm the bank's production OWD actually narrows manager vs leaf books. AWS environment + promotion gate IN PROGRESS Stand up / formalize the remote-host environment (AgentCore Runtime or ECS / Fargate in a private VPC) as the governance unit. Accenture Track 2 is already standing one up.

THE CLOSE, ON THE THROUGH-LINE

Every MCP server we bring online is one more sense for the same brain: deal-level underwriting → relationship-level monitoring → beyond, built step by step, with the policy fence designed in before the first tool call, and a path to enterprise-bank scale on the bank's own AWS.